

Unit

3

Lesson

2



Beyond Imagination.

Say It, Code It!

Time Frame: [90 min]

Lesson Objectives:

By the end of this lesson students will be able to:

- Use PictoBlox to code a sprite that responds to voice commands.
- Apply speech recognition and text-to-speech blocks in a coding activity.
- Recognize the importance of loops in programming repetitive actions.

Challenge questions of the lesson

1. How can I make a sprite in PictoBlox respond to what I say using voice commands?
2. How can I code my sprite to speak back to me using text-to-speech blocks?
3. Why are loops important when I want my sprite to repeat an action many times?

SEL Relation

- **Self-Awareness:** Recognize excitement and confidence in using coding to help others.
- **Self-Management:** Practice patience and persistence while coding and debugging.
- **Social Awareness:** Develop empathy by understanding how coding can assist people like Uncle Sam.
- **Relationship Skills:** Communicate clearly and collaborate with classmates during coding activities.
- **Responsible Decision-Making:** Make thoughtful, ethical choices when creating programs to help others.

Engage



Teacher's Role

- Set the stage with an engaging story conversation between Ms. Sara and the students to spark curiosity about using code to help others.
- Use expressive reading to introduce the idea of voice-command coding in a relatable, real-world context.
- Guide students in making personal connections by prompting them to imagine using code to help someone at home.
- Support students as they respond to story-based reflection questions, ensuring they understand that coding can solve real-life problems.

Expected From Students

- Actively listen to the story and think about how voice command coding could help someone like Uncle Sam.
- Reflect on how they would feel about using coding to help others by answering emotion-based questions.
- Identify practical examples of coding for assistance at home by selecting the correct image (e.g., turning on the lights with their voice).
- Begin connecting coding with empathy, real-world problem-solving, and helping others in daily life.

Parts of the books included.



Story



- Begin by saying:
 - “Today, we’re going to use our coding skills to help people—just like real computer scientists! Let’s start with a short story about Lila and Ms. Sara.”
- Read the dialogue aloud using different voices for Ms. Sara and Lila to make the story engaging and easy to follow.
- After reading, ask the class:
 - “What do you think voice command means?”
 - “Who do you think Uncle Sam is, and how could this help him?”
- Encourage curiosity by saying:
 - “We’re going to try coding something just like this today!”

Possible Strategies for Implementation:

1. Turn and Talk

- After reading, have students turn to a partner and discuss:
 - “How could voice control help someone in real life?”

2. Think-Pair-Share

- Ask students to quietly think about a time they used technology to help someone, then share with a partner, and finally with the class.

3. See-Think-Wonder (Visible Thinking Routine)

- Show a picture of a smart speaker or voice-controlled device.
- Ask:
 - What do you see?
 - What do you think is happening?
 - What do you wonder about how this works?

4. Anchor Vocabulary

- Introduce keywords: voice command, code blocks, AI helper.
- Write them on the board and ask students to help define them in their own words after the story.

5. Sketch to Reflect

- Ask students to draw a picture of what they think “using voice to turn on the light” would look like.

Story Reflection



Objective:

Students will reflect on the emotional and practical impact of using coding to help others. They will demonstrate understanding of how coding, particularly voice commands, can be applied in real-life home situations to support others.

Answers:

How would you feel if you could use coding to help someone at home?

(Any answer is valid—this question gauges emotional connection and self-awareness.)

Which of these is a way coding could help someone at home?

Correct Answer: Turn on the lights with their voice.

Incorrect Answers: Make a sandwich, Draw a picture, Read a storybook

(These actions are not directly supported by simple coding or AI as introduced in the lesson.)

Possible Strategies for Implementation:

1. Think-Pair-Share

- Have students choose their feeling (Question 1), share it with a partner, and explain why they chose it.

2. Class Vote

- Use hand-raising or stickers to collect how many students chose each feeling, and talk briefly about what it tells us.

3. Picture Sorting

- Print and cut out the image options for Question 2 and have students sort them into “AI could help with this” and “AI cannot help with this.”

4. Connection Circle

- Invite a few students to share:
 - “If I could use voice commands at home, I would use it to help...”

5. Draw to Show Understanding

- Let students draw their own idea of how they could use coding at home to help someone—this reinforces conceptual application.

Let's Think!



Objective:

Students will identify how to use voice command and coding blocks in PictoBlox to make a sprite respond to speech, demonstrating understanding of basic AI concepts such as speech recognition and action-triggering in a digital environment.

Answers:

Correct Answers:

- Use coding blocks to make the lamp sprite turn on when we say “Lights on.”
- Add a voice command block and connect it to the lamp sprite block.
- Use blocks that tell the sprite what to do when it hears a word.
- Say “Turn on” and the sprite follows the code we made.

Incorrect/Trick Options:

- Shout very loud at the computer until it works.
- Use coloring blocks to decorate the lamp.
- Connect wires to the lamp sprite.
- Put stickers on the sprite to make it smart.

Possible Strategies for Implementation:

1. Read Aloud and Eliminate

- Read each choice aloud and let students vote “Yes” or “No” by thumbs up/down. Cross out silly or incorrect ones together to reinforce understanding.

2. Partner Sorting

- Provide printed cutouts of each option and let students work in pairs to sort them into “Makes sense in coding” vs. “Not how coding works.”

3. Think-Pair-Share

- Ask students:
 - “Why do you think this block would work?”
 - or
 - “Which one of these would really help the sprite respond to your voice?”

4. Anchor Chart Connection

- Link correct answers to an anchor chart titled “What Makes a Smart Sprite?” showing how speech recognition and code blocks work together.

Explorer's Toolkit



Teacher's Role

- Introduce the activity by explaining that students will help an artist draw shapes using coding blocks.
- Demonstrate how to start the level by accessing the link:
<https://studio.code.org/courses/coursesec-2025/units/1/lessons/6/levels/1>
- Guide students in understanding how commands control the artist's movement on the screen.
- Provide support as students experiment with directional and repeat blocks to draw accurately.
- Pause for reflection by asking: "What did you notice about how the code made the shape?"



Expected From Students

- Open the Code.org activity and begin exploring how to move the artist using coding blocks.
- Experiment with different block sequences to draw specific shapes.
- Identify patterns in how repeated movements (like turns or steps) create lines or curves.
- Begin to recognize the value of loops or repeated instructions when drawing similar parts.
- Stay engaged, try different strategies, and ask for help when needed.

Parts of the books included

Explorer's Toolkit



DIY Lab

Let's play a fun game and help the artist draw the right shapes!

<https://studio.code.org/courses/course2/units/1/lessons/7/levels/1>



DIY Lab

Objective:

Students will use basic block-based coding to direct a digital artist to draw specific shapes. They will practice sequencing, direction, and repetition through a fun and interactive coding game.

Materials Needed:

- Devices with internet access (laptop or tablet)
- Headphones (optional for sound support)
- Student login (if saving progress)

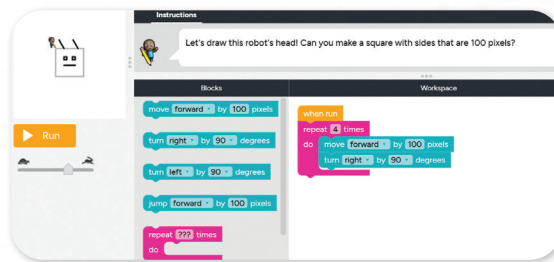
Access to the activity link:

<https://studio.code.org/courses/coursesec-2025/units/1/lessons/6/levels/1>



Activity Instructions:

1. Open the Code.org activity using the provided link.
2. Read the directions and help the artist move by snapping blocks together in the right order.
3. Use motion and turn blocks to guide the artist to draw lines, angles, and shape



4. Try each level, correcting mistakes by adjusting the sequence or directions.
5. Watch how the artist follows your instructions and draws based on your code.

Note: Adjust the number of levels based on the time available for the session, but ensure that students complete at least two levels.

Possible Strategies for Implementation:

- **Demonstrate First**
 - Show the first level on a projector or screen and solve it as a class to model how block placement works.
- **Pair Programming**
 - Have students work in pairs—one navigates, one places blocks. Then switch roles after each level.
- **Verbal Coding**
 - Before coding, ask students to describe out loud the steps the artist needs to take, reinforcing logical sequencing.
- **Debugging Practice**
 - Encourage students to test and revise their code, reminding them that fixing mistakes is part of the process.
- **Shape Hunt**
 - After completing a few levels, challenge students to draw their own shape on paper and think about what blocks would recreate it in the game.



Teacher's Role:

- Introduce the video by briefly explaining what speech recognition is (e.g., “It’s how computers understand and respond to our voice!”).
- Pause at key moments to explain or ask guiding questions like:
 1. “What’s one way speech recognition helps people?”
 2. “Have you ever talked to a smart speaker or voice assistant?”
- Clarify any vocabulary (e.g., speech, recognize, command).
- Facilitate a quick discussion after the video to reinforce understanding and make real-world connections.

Expected From Students:

- Watch the video attentively to learn how speech recognition works.
- Identify ways speech recognition is used in real life (e.g., smart home devices, voice search, accessibility tools).
- Share one idea from the video about how speech recognition can help people.
- Begin to understand how this technology connects to the coding activity they will try later (e.g., making a sprite respond to voice).

Parts of the books included.



Screen Lessons

Click on the link to learn about speech recognitions.

<https://youtu.be/LJVgBy9MEQM?feature=shared>



Screen Lessons

Objective:

Students will understand the concept of speech recognition and how it allows computers to respond to voice commands. They will explore how this technology is used in everyday life and begin making connections to coding their own voice-responsive sprite.

Video Link: <https://youtu.be/LJVgBy9MEQM?feature=shared>



Possible Strategies for Implementation:

1. Pre-Watching – KWL Chart

- Begin with a quick KWL chart:
 - K: What do you already know about computers listening to people?
 - W: What do you want to know about how it works?

2. During Watching – Pause and Prompt

- Pause at key points to ask guiding questions like:
 - “What is speech recognition used for?”
 - “Can you think of something at home that listens to your voice?”

3. Post-Watching – Think-Pair-Share

- Ask students to think:
 - “How could speech recognition help someone who can’t move easily?”
 - Let them discuss with a partner, then share with the class.
- Visual Recap
 - Draw or display a diagram showing a person speaking → computer hears → computer acts (e.g., turns on light).
- Connect to Coding
 - Remind students: “We’ll use these ideas to make our own sprite respond to our voice using code!”

Elaborate and Evaluate



Teacher’s Role

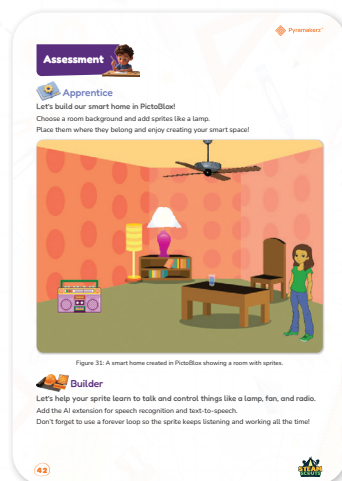
- Guide students through designing a smart home scene in PictoBlox by choosing a background and adding functional sprites (e.g., lamp, fan, radio).
- Support students in applying the AI extension for speech recognition and text-to-speech functions.
- Model or demonstrate how to use a forever loop to keep the sprite listening for commands.
- Circulate during the activity to ask guiding questions like:
 1. “What happens when you say the command?”
 2. “Is your sprite listening the whole time?”
- Encourage creativity while ensuring students are correctly applying blocks and coding logic.
- Use observation or a checklist to assess how well students set up their smart environment and program voice interactions.

Expected From Students

- Use PictoBlox to create a smart home scene by choosing a room background and adding relevant sprites (e.g., lamp, radio).
- Arrange sprites logically in the scene to reflect a real room layout.
- Apply speech recognition and text-to-speech blocks to make the sprite respond to voice commands.

- Use a forever loop to keep their sprite active and responsive throughout the activity.
- Test their code and revise as needed to ensure their sprite works as a smart helper.
- Demonstrate understanding of coding logic, voice input, and smart system control by showcasing their working smart scene.

Parts of the books included.



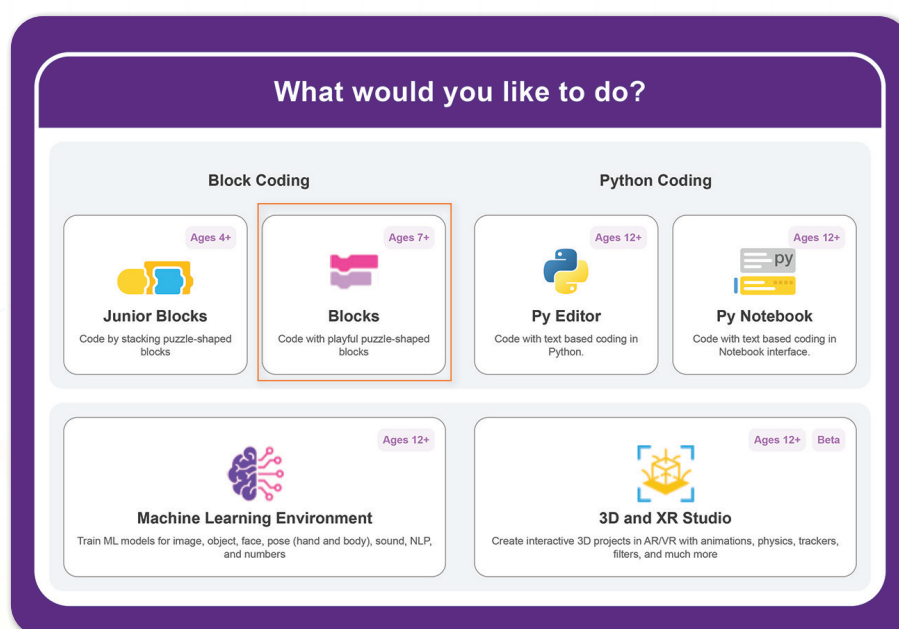
Apprentice

Objective:

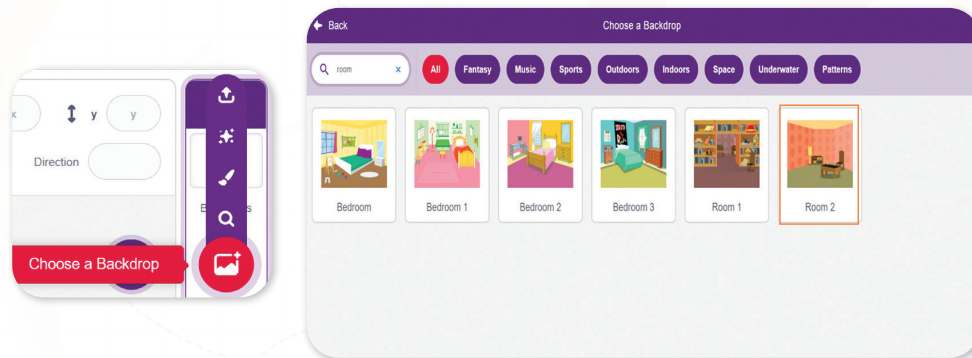
Students will design a digital smart home scene in PictoBlox by selecting a room background and placing functional sprites (like a lamp or fan) in logical locations. This activity helps develop spatial awareness, scene composition, and prepares students to later program interactive smart features.

Activity Instructions:

1. Open PictoBlox and start a new project.

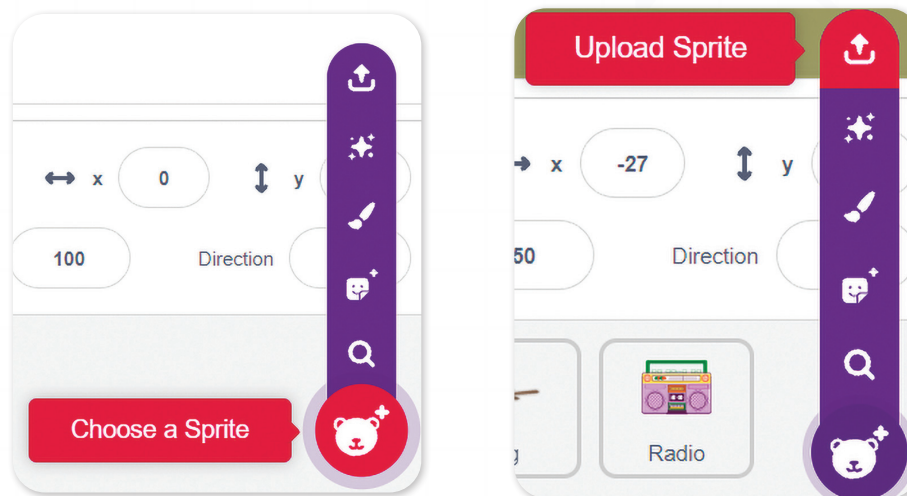


2. Choose a background that looks like a room in a house (e.g., bedroom, living room).

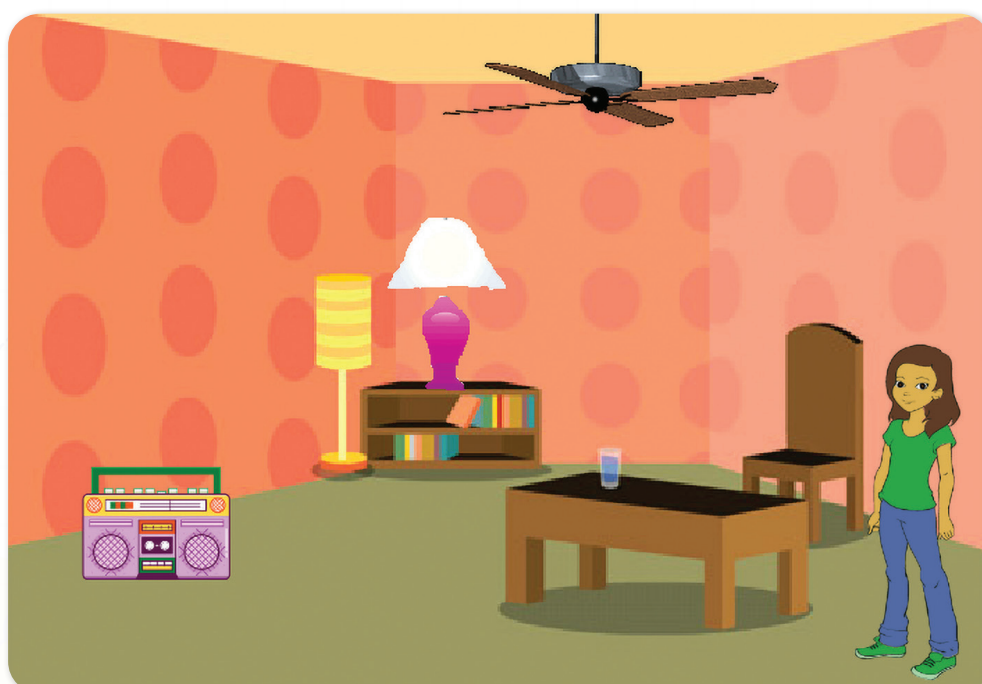


3. Insert sprites that represent smart devices commonly found in a smart home (such as a lamp, fan, speaker, or person).

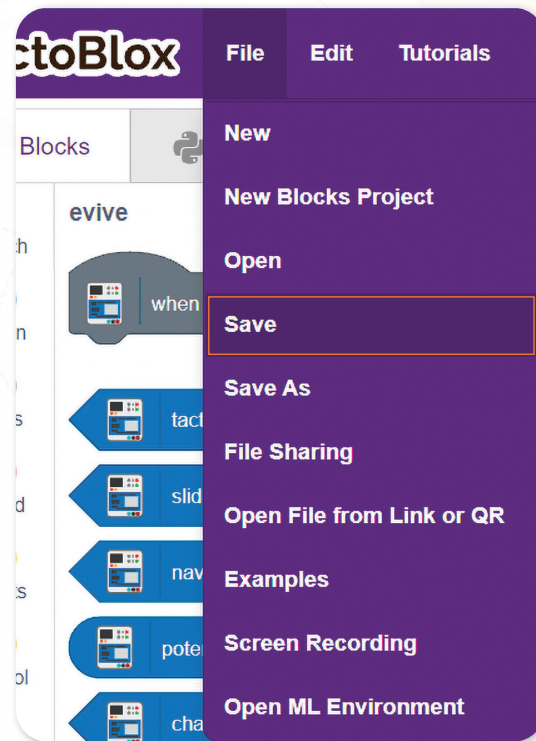
Note: These specific items aren't available by default in the program, so you'll need to download them from the following link: https://drive.google.com/drive/folders/1K3x_w6sSEV3g9Xztb9ZUblm7I2DJ9V6L?usp=drive_link



4. Place the sprites in appropriate places within the room.



5. Use your imagination to create a welcoming, smart space.
6. Save your work—you'll add code to these sprites later to make them “smart.”



Possible Strategies for Implementation:

- **Model First:** Show students a quick demo of how to choose a background and drag sprites into place in PictoBlox.
- **Spatial Reasoning Prompts:** Ask questions like, “Where would the lamp go?” or “Why would you put the speaker there?”
- **Partner Feedback:** Allow students to show their layout to a partner and explain their choices before coding begins.
- **Design Challenge:** Encourage students to include at least 3 smart items and place them in different parts of the room.
- **Vocabulary Connection:** Highlight terms like “sprite,” “background,” and “scene” to reinforce technical language in context.



Builder

Objective:

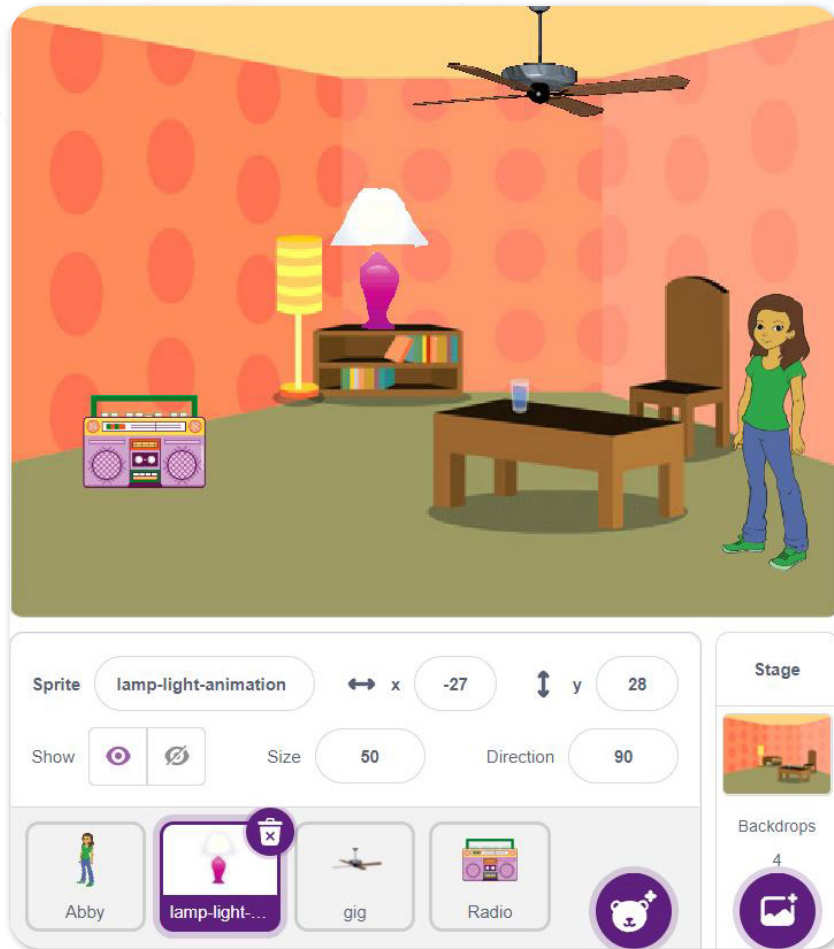
Students will use PictoBlox to program a sprite that responds to voice commands using speech recognition and text-to-speech. They will apply loops to keep the sprite listening and functioning continuously, simulating smart device behavior.

Project link: https://drive.google.com/file/d/1CBH4MXLbnAxn2pi4Sj9sLWaIMJIntDT/view?usp=drive_link

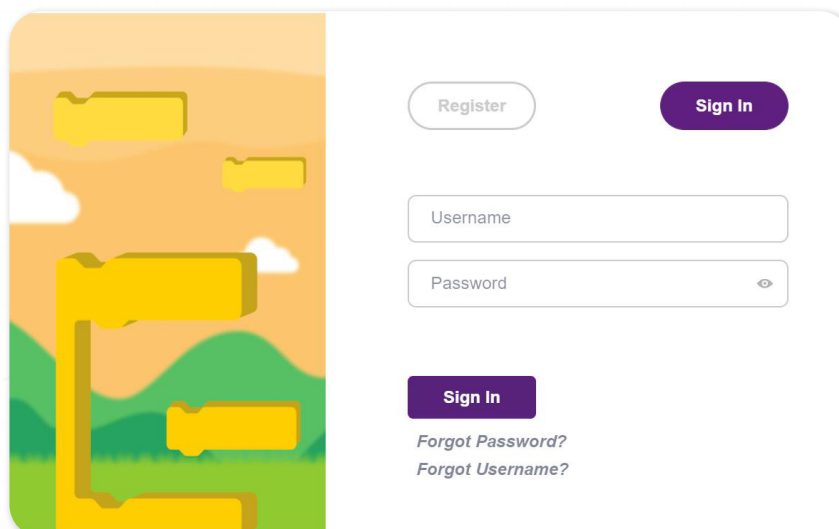


Activity Instructions

1. Open your smart home project in PictoBlox.

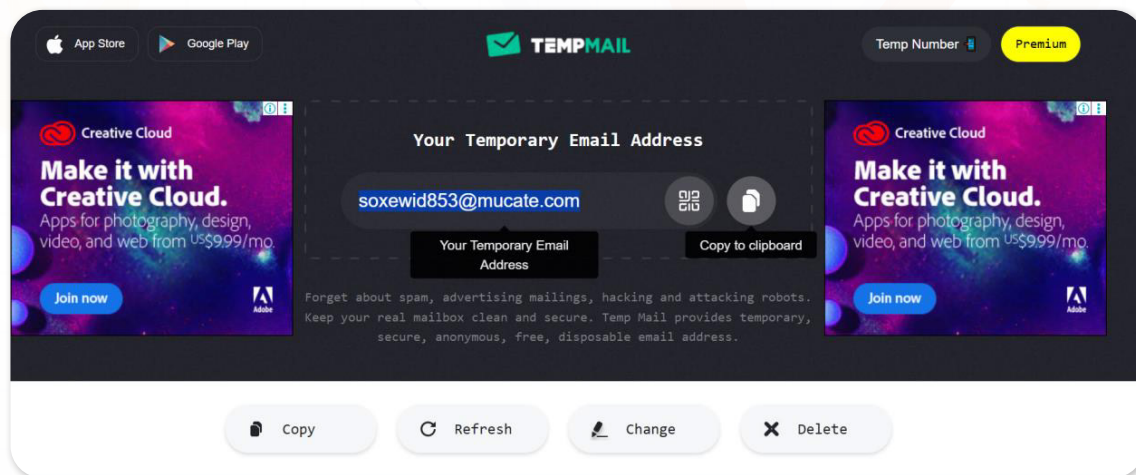


2. Sign in in pictoblox to use AI features:

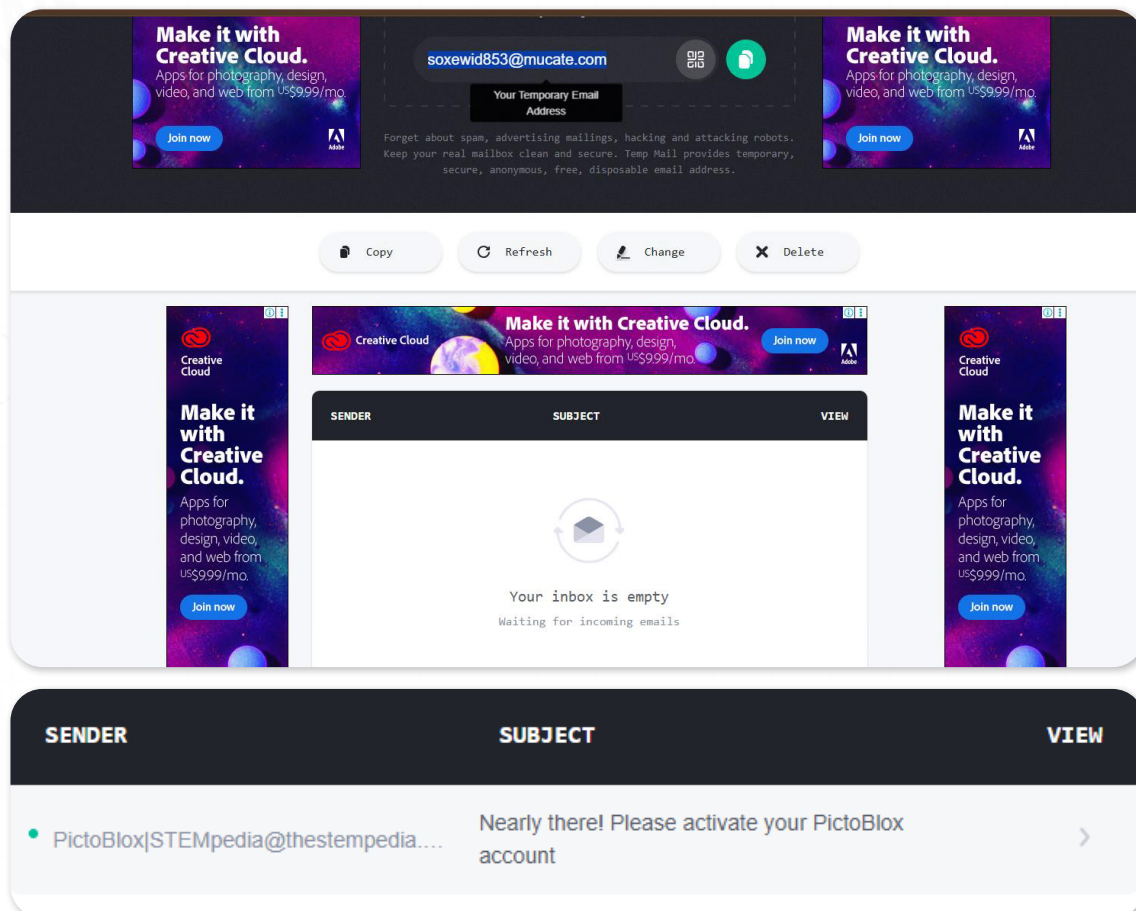


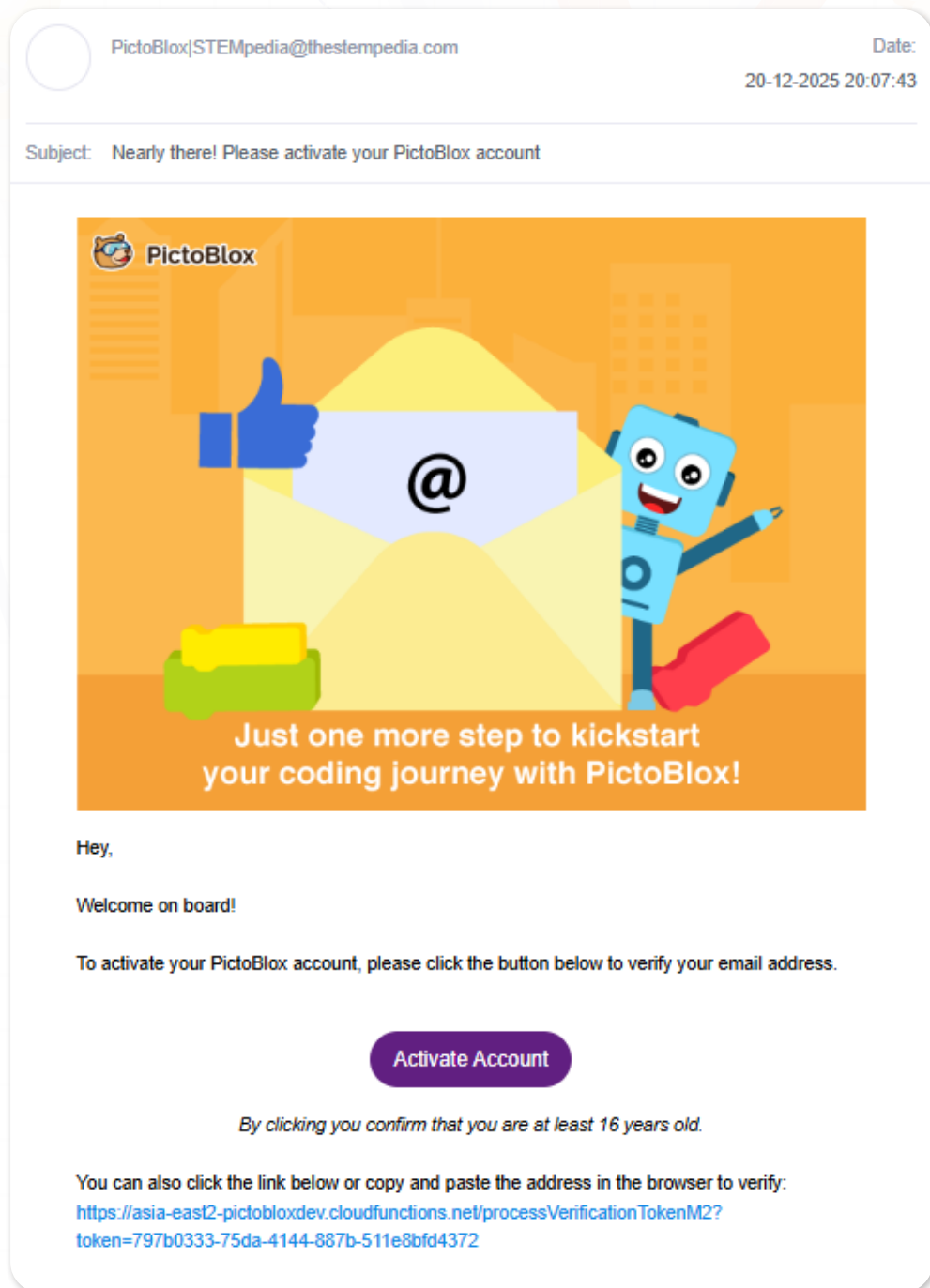
Note: Each account has a limited credit, and once it is used up, the AI features stop working. In this case, we can use Temp Mail (<https://temp-mail.org/>) to quickly generate an email address for signing in.



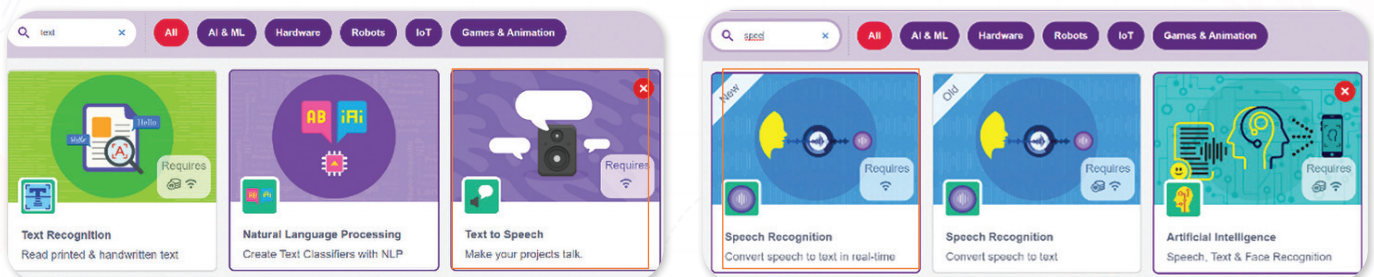


- After signing up, you will receive a verification email, which can be found in the Temp Mail inbox.

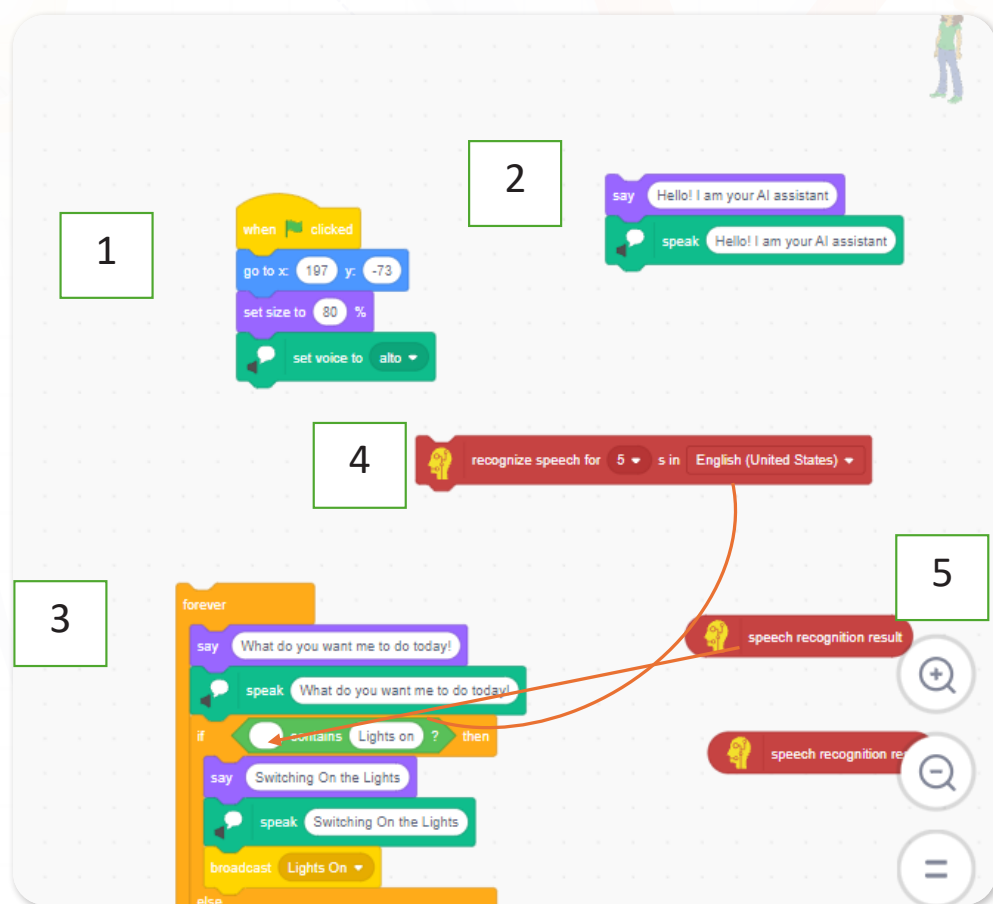




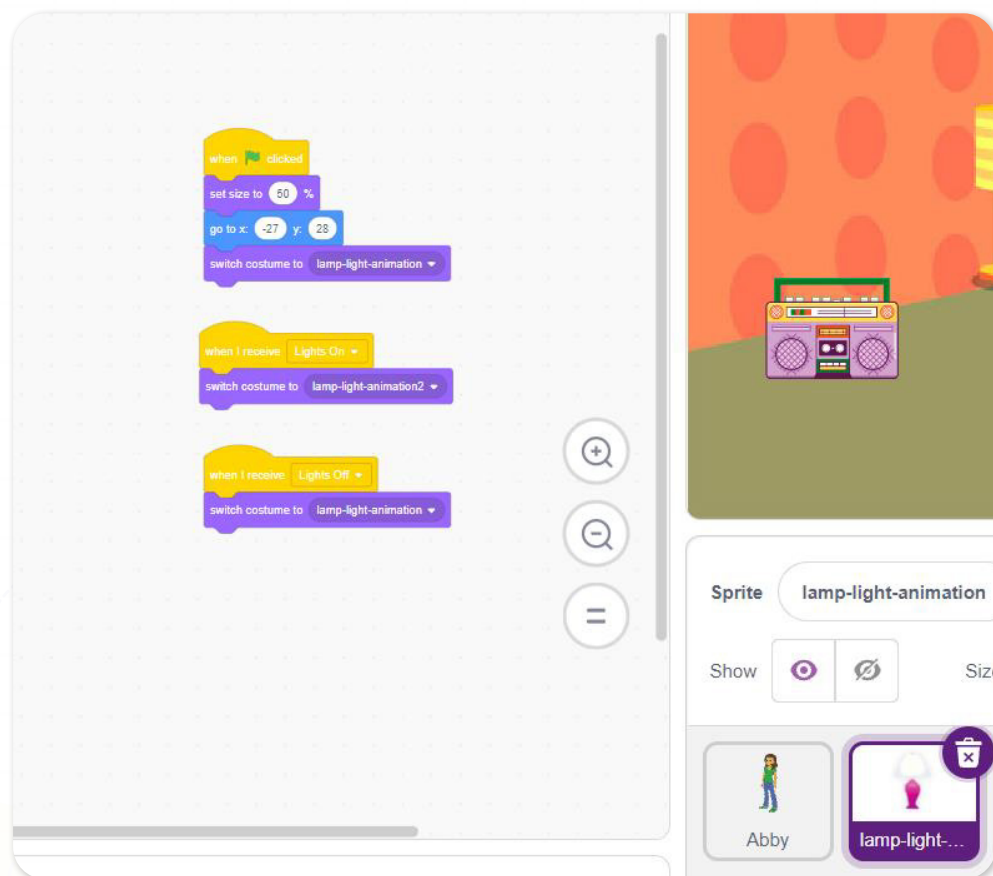
- Back to pictoblox: Add the AI Extension for Speech Recognition and Text-to-Speech.

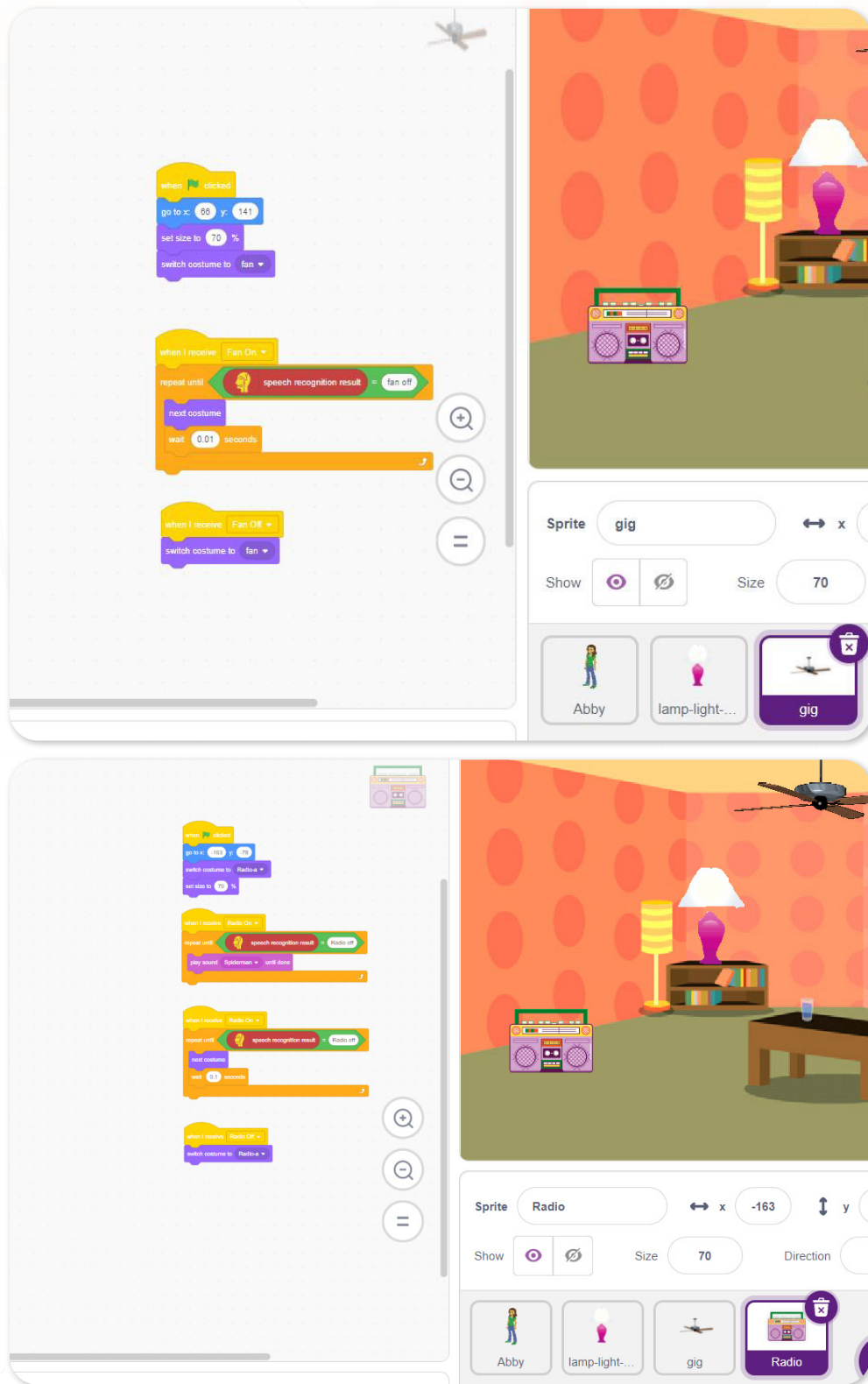


- Do not ask students to write the entire code from scratch. Instead, provide them with files containing scattered code for the Abby sprite and have them organize it, with a focus on the speech recognition sprites.



6. Let the students code the three other sprites from scratch.





7. Test and adjust your code to make sure the sprite follows your voice instructions correctly.

Rubric: Smart Home Voice-Controlled Sprite.

Criteria	Excellent (5 pts)	Good (4 pts)	Developing (3 pts)	Needs Improvement (1–2 pts)
Speech Recognition Setup	Correctly adds and configures speech recognition to detect specific commands	Adds speech recognition but commands are too broad or not always detected	Attempts to use speech recognition but not functioning correctly	Does not include or configure the speech recognition extension
Text-to-Speech Integration	Sprite clearly responds using text-to-speech and says meaningful responses	Sprite responds but speech is unclear or not well-matched to the voice input	Speech is triggered but lacks clarity or relevance	No speech response implemented
Loop Usage	Uses a forever loop effectively so sprite listens continuously	Loop is used but doesn't fully maintain continuous listening	Loop is included but incorrectly placed or limited	No loop used or not working properly
Code Functionality and Logic	Program works smoothly with logical flow and accurate responses	Program works but has minor issues in flow or logic	Program runs inconsistently or partially	Code has major bugs or doesn't respond to input
Creativity and Effort	Adds extra sprites, creative voice commands, and a well-decorated smart space	Adds appropriate sprites and background with some customization	Basic setup with minimal customization	Lacks effort in design; minimal or default sprites and setup

It's time to move to the create part and add this.

Create

We started building our smart home using PictoBlox. We picked a room background and added sprites like a lamp, fan, and radio. Then we used blocks to help the sprite listen to our voice. We added the speech recognition and text-to-speech extensions so the sprite could understand what we said and talk back to us. We made sure to use a forever loop so it keeps listening all the time. Our goal was to help people—just like Uncle Sam—by making a smart home that responds to voice commands and makes life easier.



DIY Lab

Objective:

Students will reflect on their learning by identifying key coding and AI concepts used while creating a voice-controlled smart home scene. They will demonstrate understanding of speech recognition, text-to-speech, and loops by describing how their sprite responds to voice, talks back, and keeps running using a loop block.

Activity Instructions

1. Review Each “Now I Can” Statement Together:
 - Read each statement aloud and ask students to share examples from their PictoBlox project (e.g., “What did your sprite do when you said ‘Turn on the lamp?’”).
2. Encourage Self-Assessment:
 - Give students a copy of the “Now I Can” checklist. Ask them to check off each skill they used while designing and coding their smart home sprite.

Relation to Standards

1. ISTE Standards for Students

1.1 Empowered Learner:

Students use technology to set goals and demonstrate competency by creating interactive smart home projects that respond to voice input.

1.4 Innovative Designer:

Students apply the design process to develop creative solutions for everyday challenges using coding and AI tools.

1.5 Computational Thinker:

Students build and test logical code using loops, speech recognition, and conditionals to make sprites act based on voice commands.

2. CSTA K–12 Computer Science Standards

1B-AP-10:

Create programs that include sequences, events, loops, and conditionals.

1B-AP-15:

Test and debug (identify and fix errors) a program or algorithm to ensure it runs as intended.

1B-IC-21:

Discuss how computing technologies can help people (e.g., smart homes for accessibility).

SDG 9: Industry, Innovation, and Infrastructure

Students explore how smart technologies like AI and voice control can be designed to improve quality of life, especially for people with disabilities.